

INTERNATIONAL STANDARD

**Industrial communication networks – Fieldbus specifications –
Part 6-16: Application layer protocol specification – Type 16 elements**



THIS PUBLICATION IS COPYRIGHT PROTECTED

Copyright © 2007 IEC, Geneva, Switzerland

All rights reserved. Unless otherwise specified, no part of this publication may be reproduced or utilized in any form or by any means, electronic or mechanical, including photocopying and microfilm, without permission in writing from either IEC or IEC's member National Committee in the country of the requester.

If you have any questions about IEC copyright or have an enquiry about obtaining additional rights to this publication, please contact the address below or your local IEC member National Committee for further information.

IEC Central Office
3, rue de Varembe
CH-1211 Geneva 20
Switzerland
Email: inmail@iec.ch
Web: www.iec.ch

About the IEC

The International Electrotechnical Commission (IEC) is the leading global organization that prepares and publishes International Standards for all electrical, electronic and related technologies.

About IEC publications

The technical content of IEC publications is kept under constant review by the IEC. Please make sure that you have the latest edition, a corrigenda or an amendment might have been published.

- Catalogue of IEC publications: www.iec.ch/searchpub

The IEC on-line Catalogue enables you to search by a variety of criteria (reference number, text, technical committee,...). It also gives information on projects, withdrawn and replaced publications.

- IEC Just Published: www.iec.ch/online_news/justpub

Stay up to date on all new IEC publications. Just Published details twice a month all new publications released. Available on-line and also by email.

- Electropedia: www.electropedia.org

The world's leading online dictionary of electronic and electrical terms containing more than 20 000 terms and definitions in English and French, with equivalent terms in additional languages. Also known as the International Electrotechnical Vocabulary online.

- Customer Service Centre: www.iec.ch/webstore/custserv

If you wish to give us your feedback on this publication or need further assistance, please visit the Customer Service Centre FAQ or contact us:

Email: csc@iec.ch
Tel.: +41 22 919 02 11
Fax: +41 22 919 03 00



IEC 61158-6-16

Edition 1.0 2007-12

INTERNATIONAL STANDARD

**Industrial communication networks – Fieldbus specifications –
Part 6-16: Application layer protocol specification – Type 16 elements**

INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

PRICE CODE

T

ICS 35.100.70; 25.040.40

ISBN 2-8318-9494-8

LICENSED TO MECON Limited, - RANCHI/BANGALORE
FOR INTERNAL USE AT THIS LOCATION ONLY, SUPPLIED BY BOOK SUPPLY BUREAU.

CONTENTS

FOREWORD	4
INTRODUCTION	6
1 Scope	7
1.1 General	7
1.2 Specifications	8
1.3 Conformance	8
2 Normative references	8
3 Terms, definitions, abbreviations, symbols and conventions	8
3.1 Referenced terms and definitions	8
3.2 Additional terms and definitions for Type 16	10
3.3 Additional abbreviations and symbols for Type 16	11
3.4 Conventions	11
4 Abstract syntax	11
5 Transfer syntax	12
5.1 Introduction	12
5.2 RTC-MDT PDU merged abstract and transfer syntax	12
5.3 RTC-AT PDU merged abstract and transfer syntax	12
5.4 RTC-MDT PDU encoding	13
5.5 RTC-AT PDU encoding	13
6 Structure of FAL protocol state machines	13
7 AP-Context state machine	15
7.1 Overview	15
7.2 States	15
7.3 States, events and transitions	15
8 FAL service protocol machine (FSPM)	16
8.1 Overview	16
8.2 MGT services	16
8.3 IDN services	17
8.4 CYCIDN services	17
8.5 File transmission services	17
9 Application relationship protocol machine (ARPM)	18
9.1 Overview	18
9.2 Master ARPM	18
9.3 Slave ARPM	19
9.4 Primitives received from the FSPM	20
9.5 Indications received from the DMPM	22
10 DLL mapping protocol machine (DMPM)	23
10.1 Overview	23
10.2 Primitives received from the ARPM	23
10.3 Indications received from the DL	23
Bibliography	24
Figure 1 – Relationships among protocol machines and adjacent layers	14
Figure 2 – APCSM state diagram	15
Figure 3 – ARPM master AR state diagram	18

Figure 4 – ARPM slave AR state diagram	19
Table 1 – RTC-MDT PDU attribute format	12
Table 2 – RTC-AT PDU attribute format	12
Table 3 – RTC-MDT PDU attribute encoding	13
Table 4 – RTC-AT PDU attribute encoding	13
Table 5 – APCSM state-event table	16
Table 6 – ARPM master state-event table	19
Table 7 – ARPM slave state-event table	20
Table 8 – ARPM to DL mapping	23
Table 9 – DL to ARPM mapping	23

INTERNATIONAL ELECTROTECHNICAL COMMISSION

INDUSTRIAL COMMUNICATION NETWORKS – FIELD BUS SPECIFICATIONS –

Part 6-16: Application layer protocol specification – Type 16 elements

FOREWORD

- 1) The International Electrotechnical Commission (IEC) is a worldwide organization for standardization comprising all national electrotechnical committees (IEC National Committees). The object of IEC is to promote international co-operation on all questions concerning standardization in the electrical and electronic fields. To this end and in addition to other activities, IEC publishes International Standards, Technical Specifications, Technical Reports, Publicly Available Specifications (PAS) and Guides (hereafter referred to as "IEC Publication(s)"). Their preparation is entrusted to technical committees; any IEC National Committee interested in the subject dealt with may participate in this preparatory work. International, governmental and non-governmental organizations liaising with the IEC also participate in this preparation. IEC collaborates closely with the International Organization for Standardization (ISO) in accordance with conditions determined by agreement between the two organizations.
- 2) The formal decisions or agreements of IEC on technical matters express, as nearly as possible, an international consensus of opinion on the relevant subjects since each technical committee has representation from all interested IEC National Committees.
- 3) IEC Publications have the form of recommendations for international use and are accepted by IEC National Committees in that sense. While all reasonable efforts are made to ensure that the technical content of IEC Publications is accurate, IEC cannot be held responsible for the way in which they are used or for any misinterpretation by any end user.
- 4) In order to promote international uniformity, IEC National Committees undertake to apply IEC Publications transparently to the maximum extent possible in their national and regional publications. Any divergence between any IEC Publication and the corresponding national or regional publication shall be clearly indicated in the latter.
- 5) IEC provides no marking procedure to indicate its approval and cannot be rendered responsible for any equipment declared to be in conformity with an IEC Publication.
- 6) All users should ensure that they have the latest edition of this publication.
- 7) No liability shall attach to IEC or its directors, employees, servants or agents including individual experts and members of its technical committees and IEC National Committees for any personal injury, property damage or other damage of any nature whatsoever, whether direct or indirect, or for costs (including legal fees) and expenses arising out of the publication, use of, or reliance upon, this IEC Publication or any other IEC Publications.
- 8) Attention is drawn to the Normative references cited in this publication. Use of the referenced publications is indispensable for the correct application of this publication.
- 9) Attention is drawn to the possibility that some of the elements of this IEC Publication may be the subject of patent rights. IEC shall not be held responsible for identifying any or all such patent rights.

NOTE Use of some of the associated protocol types is restricted by their intellectual-property-right holders. In all cases, the commitment to limited release of intellectual-property-rights made by the holders of those rights permits a particular data-link layer protocol type to be used with physical layer and application layer protocols in Type combinations as specified explicitly in the IEC 61784 series. Use of the various protocol types in other combinations may require permission from their respective intellectual-property-right holders.

International Standard IEC 61158-6-16 has been prepared by subcommittee 65C: Industrial networks, of IEC technical committee 65: Industrial-process measurement, control and automation.

This first edition and its companion parts of the IEC 61158-6 subseries cancel and replace IEC 61158-6:2003. This edition of this part constitutes a technical addition. This publication, together with its companion parts for Type 16, also partially replaces IEC 61491:2002 which is at present being revised. IEC 61491 will be issued as a technical report.

This edition of IEC 61158-6 includes the following significant changes from the previous edition:

- a) deletion of the former Type 6 fieldbus for lack of market relevance;

- b) addition of new types of fieldbuses;
- c) partition of part 6 of the third edition into multiple parts numbered -6-2, -6-3, ...

The text of this standard is based on the following documents:

FDIS	Report on voting
65C/476/FDIS	65C/487/RVD

Full information on the voting for the approval of this standard can be found in the report on voting indicated in the above table.

This publication has been drafted in accordance with ISO/IEC Directives, Part 2.

The committee has decided that the contents of this publication will remain unchanged until the maintenance result date indicated on the IEC web site under <http://webstore.iec.ch> in the data related to the specific publication. At this date, the publication will be:

- reconfirmed;
- withdrawn;
- replaced by a revised edition, or
- amended.

NOTE The revision of this standard will be synchronized with the other parts of the IEC 61158 series.

The list of all the parts of the IEC 61158 series, under the general title *Industrial communication networks – Fieldbus specifications*, can be found on the IEC web site.

INTRODUCTION

This part of IEC 61158 is one of a series produced to facilitate the interconnection of automation system components. It is related to other standards in the set as defined by the “three-layer” fieldbus reference model described in IEC/TR 61158-1.

The application protocol provides the application service by making use of the services available from the data-link or other immediately lower layer. The primary aim of this standard is to provide a set of rules for communication expressed in terms of the procedures to be carried out by peer application entities (AEs) at the time of communication. These rules for communication are intended to provide a sound basis for development in order to serve a variety of purposes:

- as a guide for implementors and designers;
- for use in the testing and procurement of equipment;
- as part of an agreement for the admittance of systems into the open systems environment;
- as a refinement to the understanding of time-critical communications within OSI.

This standard is concerned, in particular, with the communication and interworking of sensors, effectors and other automation devices. By using this standard together with other standards positioned within the OSI or fieldbus reference models, otherwise incompatible systems may work together in any combination.

INDUSTRIAL COMMUNICATION NETWORKS – FIELDBUS SPECIFICATIONS –

Part 6-16: Application layer protocol specification – Type 16 elements

1 Scope

1.1 General

The fieldbus Application Layer (FAL) provides user programs with a means to access the fieldbus communication environment. In this respect, the FAL can be viewed as a “window between corresponding application programs.”

This standard provides common elements for basic time-critical and non-time-critical messaging communications between application programs in an automation environment and material specific to Type 16 fieldbus. The term “time-critical” is used to represent the presence of a time-window, within which one or more specified actions are required to be completed with some defined level of certainty. Failure to complete specified actions within the time window risks failure of the applications requesting the actions, with attendant risk to equipment, plant and possibly human life.

This standard define in an abstract way the externally visible service provided by the different Types of fieldbus Application Layer in terms of

- an abstract model for defining application resources (objects) capable of being manipulated by users via the use of the FAL service,
- the primitive actions and events of the service;
- the parameters associated with each primitive action and event, and the form which they take; and
- the interrelationship between these actions and events, and their valid sequences.

The purpose of this standard is to define the services provided to

- the FAL user at the boundary between the user and the Application Layer of the Fieldbus Reference Model, and
- Systems Management at the boundary between the Application Layer and Systems Management of the Fieldbus Reference Model.

This standard specifies the structure and services of the IEC fieldbus Application Layer, in conformance with the OSI Basic Reference Model (ISO/IEC 7498) and the OSI Application Layer Structure (ISO/IEC 9545).

FAL services and protocols are provided by FAL application-entities (AE) contained within the application processes. The FAL AE is composed of a set of object-oriented Application Service Elements (ASEs) and a Layer Management Entity (LME) that manages the AE. The ASEs provide communication services that operate on a set of related application process object (APO) classes. One of the FAL ASEs is a management ASE that provides a common set of services for the management of the instances of FAL classes.

Although these services specify, from the perspective of applications, how request and responses are issued and delivered, they do not include a specification of what the requesting and responding applications are to do with them. That is, the behavioral aspects of the applications are not specified; only a definition of what requests and responses they can send/receive is specified. This permits greater flexibility to the FAL users in standardizing

such object behavior. In addition to these services, some supporting services are also defined in this standard to provide access to the FAL to control certain aspects of its operation.

1.2 Specifications

The principal objective of this standard is to specify the syntax and behavior of the application layer protocol that conveys the application layer services defined in IEC 61158-5-16.

A secondary objective is to provide migration paths from previously-existing industrial communications protocols. It is this latter objective which gives rise to the diversity of protocols standardized in subparts of IEC 61158-6.

1.3 Conformance

This standard does not specify individual implementations or products, nor does it constrain the implementations of application layer entities within industrial automation systems.

There is no conformance of equipment to this application layer service definition standard. Instead, conformance is achieved through implementation of conforming application layer protocols that fulfill any given Type of application layer services as defined in this standard.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 61158-4-16, *Industrial communication networks – Fieldbus specifications – Part 4-16: Data-link layer protocol specification – Type 16 elements*

IEC 61158-5-16, *Industrial communication networks – Fieldbus specifications – Part 5-16: Application layer service definition – Type 16 elements*

ISO/IEC 10731, *Information technology – Open Systems Interconnection – Basic Reference Model – Conventions for the definition of OSI services*

ISO/IEC 7498-1, *Information technology – Open Systems Interconnection – Basic Reference Model – Part 1: The Basic Model*

ISO/IEC 8822:1994, *Information technology – Open Systems Interconnection – Presentation service definition*

ISO/IEC 8824, *Information technology – Open Systems Interconnection – Specification of Abstract Syntax Notation One (ASN.1)*

ISO/IEC 9545, *Information technology – Open Systems Interconnection – Application Layer structure*

3 Terms, definitions, abbreviations, symbols and conventions

3.1 Referenced terms and definitions

3.1.1 ISO/IEC 7498-1 terms

For the purposes of this document, the following terms as defined in ISO/IEC 7498-1 apply:

d) application entity

- e) application process
- f) application protocol data unit
- g) application service element
- h) application entity invocation
- i) application process invocation
- j) application transaction
- k) real open system
- l) transfer syntax

3.1.2 ISO/IEC 8822 terms

For the purposes of this document, the following terms as defined in ISO/IEC 8822 apply:

- a) abstract syntax
- b) presentation context

3.1.3 ISO/IEC 9545 terms

For the purposes of this document, the following terms as defined in ISO/IEC 9545 apply:

- a) application-association
- b) application-context
- c) application context name
- d) application-entity-invocation
- e) application-entity-type
- f) application-process-invocation
- g) application-process-type
- h) application-service-element
- i) application control service element

3.1.4 ISO/IEC 8824 terms

For the purposes of this document, the following terms as defined in ISO/IEC 8824 apply:

- a) object identifier
- b) type

3.1.5 Fieldbus Data Link Layer terms

For the purposes of this document, the following terms as defined in IEC 61158-3-3 and IEC 61158-4-3 apply.

- a) DL-Time
- b) DL-Scheduling-policy
- c) DLCEP
- d) DLC
- e) DL-connection-oriented mode
- f) DLPDU
- g) DLSDU
- h) DLSAP
- i) fixed tag
- j) generic tag

- k) link
- l) MAC ID
- m) network address
- n) node address
- o) node
- p) tag
- q) scheduled
- r) unscheduled

3.2 Additional terms and definitions for Type 16

For the purposes of this document, the following terms and definitions apply.

3.3 communication cycle

fixed time period between two master synchronization telegrams in which real-time telegrams are transmitted in the RT channel and non real-time telegrams are transmitted in the IP channel

3.4 control unit

control device (e.g., a PLC as specified in the IEC 61131 standard family)

3.5 control word

two adjacent octets inside the master data telegram containing commands for the addressed device

3.6 cycle time

duration of a communication cycle

3.7 device

a slave in the communication network, (e.g., a power drive system as defined in the IEC 61800 standard family, I/O stations as defined in the IEC 61131 standard family).

3.8 device status

four adjacent octets inside the acknowledge telegram containing status information for each device

3.9 identification number (IDN)

designation of operating data under which a data block is preserved with its attribute, name, unit, minimum and maximum input values, and the data

3.10 little endian

describes a model of memory organisation which stores the least significant octet at the lowest address, or for transfer, which transfers the lowest order octet first

3.11 master data telegram (MDT)

telegram, in which the master inserts its data

3.12**protocol**

convention about the data formats, time sequences, and error correction in the data exchange of communication systems

3.13**slave**

node, which is assigned the right to transmit by the master

3.14**status word**

two adjacent octets inside the acknowledge telegram containing status information of a device

3.15**S-0-nnnn**

designation of IDNs

3.16 Additional abbreviations and symbols for Type 16

AT	acknowledge telegram
CC	procedure command control
CC-data	cross communication between participants
IDLE	inter packet gap (see IPG)
IDN	identification number
IPOSYNC	synchronization for PDS interpolator
MDT	master data telegram
RTC	real-time channel
SERCOS	serial real-time communication system interface

3.17 Conventions**3.17.1 General concept**

The FAL is defined as a set of object-oriented ASEs. Each ASE is specified in a separate subclause. Each ASE specification is composed of three parts: its class definitions, its services, and its protocol specification. The first two are contained in IEC 61158-5-16. The protocol specification for each of the ASEs is defined in this standard.

The class definitions define the attributes of the classes supported by each ASE. The attributes are accessible from instances of the class using the Management ASE services specified in IEC 61158-5-16. The service specification defines the services that are provided by the ASE.

This standard uses the descriptive conventions given in ISO/IEC 10731.

3.17.2 Conventions for Type 16

No specific Type 16 conventions

4 Abstract syntax

The abstract syntax and the transfer syntax are merged into a fixed format that is defined in the the next clause.

5 Transfer syntax

5.1 Introduction

Type 16 transfer syntax shall be bit-coded, and therefore does not comply with usual data type specifications such as integer32 and alike.

The octet encoding shall use little endian.

5.2 RTC-MDT PDU merged abstract and transfer syntax

The merged abstract and transfer syntax for attributes belonging to this class is described in Table 1.

Table 1 – RTC-MDT PDU attribute format

Attribute	Format	Size (bits)
Control word	2 Octets, bit mapped	16
Reserved field (for future extension)	5 Bits	5
IPOSYNC: Control unit synchronization bit	1 Bit	1
Reserved field (for application profile)	2 Bits	2
Real-time control bit 2	1 Bit	1
Real-time control bit 1	1 Bit	1
Reserved field (for DLL)	6 Bits	6
Configurable part of data record	List of 2, 4 or 8 Octets	
Configured cyclic command value 1	2, 4 or 8 Octets	
Configured cyclic command value 2	2, 4 or 8 Octets	
...	...	
Configured cyclic command value n	2, 4 or 8 Octets	
NOTE n = number of configured cyclic command values. The structure and content of the configurable part of the data record is determined by the configuration list labeled IDN S-0-0024, as specified in IEC 61158-4-16, A.3.17.		

5.3 RTC-AT PDU merged abstract and transfer syntax

The merged abstract and transfer syntax for attributes belonging to this class is described in Table 2.

Table 2 – RTC-AT PDU attribute format

Attribute	Format	Size (bits)
Status word	2 Octets, bit mapped	16
Reserved field (for application profile)	8 Bits	8
Real-time status bit 2	1 Bit	1
Real-time status bit 1	1 Bit	1
Procedure command change bit	1Bit	1
Real-time valid bit	1 Bit	1
Command value processing	1 Bit	1
Reserved field (for DLL)	3 Bits	3
Configurable part of data record	List of 2, 4 or 8 Octets	

Attribute	Format	Size (bits)
Configured cyclic feedback value 1	2, 4 or 8 Octets	
Configured cyclic feedback value 2	2, 4 or 8 Octets	
...	...	
Configured cyclic feedback value n	2, 4 or 8 Octets	
NOTE n = number of configured cyclic command values. The structure and content of the configurable part of the data record is determined by the configuration list labeled IDN S-0-0016, as specified in IEC 61158-4-16, A.3.12.		

5.4 RTC-MDT PDU encoding

The specific PDU encoding for attributes belonging to this class is described in Table 3.

Table 3 – RTC-MDT PDU attribute encoding

Attribute	Encoding	
	Bits	Description
Reserved field	15-11	Reserved
IPOSYNC: Control unit synchronization bit	10	Toggle-Bit
Reserved field	9-8	Reserved
Real-time control bit 2	7	Signal indicating specific events
Real-time control bit 1	6	Signal indicating specific events
Reserved field (for DLL)	5-0	Reserved

5.5 RTC-AT PDU encoding

The specific PDU encoding for attributes belonging to this class is described in Table 4.

Table 4 – RTC-AT PDU attribute encoding

Attribute	Encoding	
	Bits	Description
Reserved field	15-8	reserved
Real-time status bit 2	7	Signal indicating specific event
Real-time status bit 1	6	Signal indicating specific event
Procedure command change bit	5	See Table 2
Real-time valid bit	4	Data has been inserted by a slave device
Command value processing	3	Data are being processed by the slave device
Reserved field (for DLL)	2-0	Reserved

6 Structure of FAL protocol state machines

Interface to FAL services and protocol machines are specified in this subclause.

The behavior of the FAL is described by three integrated protocol machines. Specific sets of these protocol machines are defined for different AREP types. The three protocol machines are: FAL Service Protocol Machine (FSPM), the Application Relationship Protocol Machine (ARPM), and the Data Link Layer Mapping Protocol Machine (DMPM). The relationships

among these protocol machines as well as primitives exchanged among them are depicted in Figure 1.

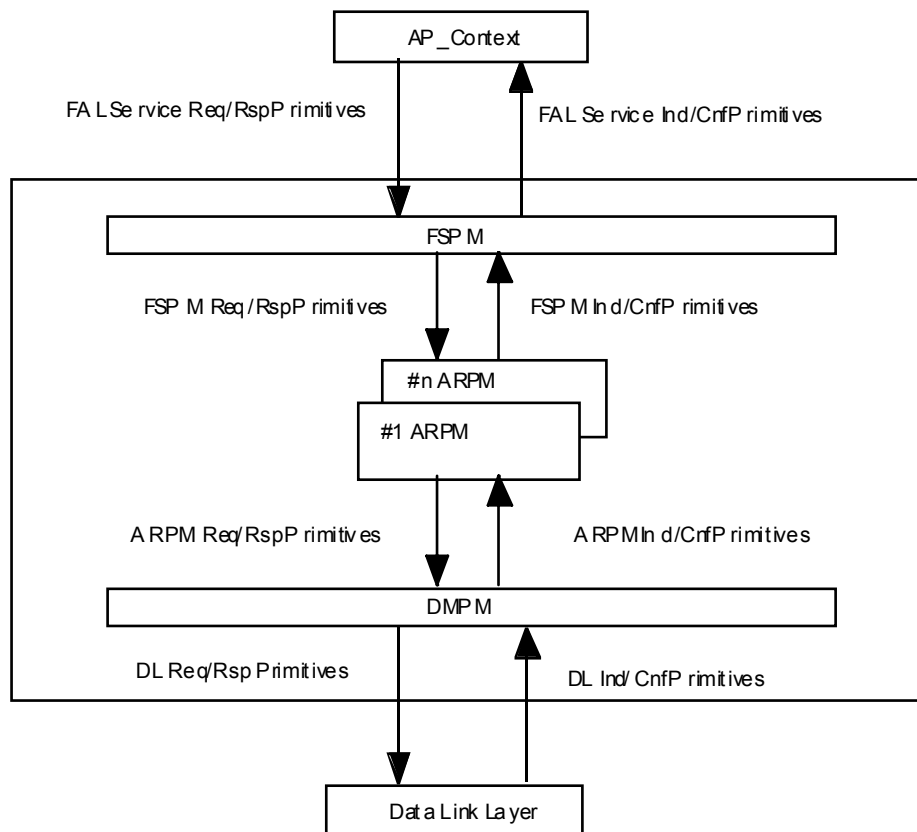


Figure 1 – Relationships among protocol machines and adjacent layers

The FSPM describes the service interface between the AP-Context and a particular AREP. The FSPM is common to all the AREP classes and does not have any state changes. The FSPM is responsible for the following activities:

- to accept service primitives from the FAL service user and convert them into FAL internal primitives;
- to select an appropriate ARPM state machine based on the AREP Identifier parameter supplied by the AP-Context and send FAL internal primitives to the selected ARPM;
- to accept FAL internal primitives from the ARPM and convert them into service primitives for the AP-Context;
- to deliver the FAL service primitives to the AP-Context based on the AREP Identifier parameter associated with the primitives.

The ARPM describes the establishment and release of an AR and exchange of FAL-PDUs with a remote ARPM(s). The ARPM is responsible for the following activities:

- to accept FAL internal primitives from the FSPM and create and send other FAL internal primitives to either the FSPM or the DMPM, based on the AREP and primitive types;
- to accept FAL internal primitives from the DMPM and send them to the FSPM as a form of FAL internal primitives;
- if the primitives are for the Establish or Abort service, it shall try to establish or release the specified AR.

The DMPM describes the mapping between the FAL and the DLL. It is common to all the AREP types and does not have any state changes. The DMPM is responsible for the following activities:

- a) to accept FAL internal primitives from the ARPM, prepare DLL service primitives, and send them to the DLL;
- b) to receive DLL indication or confirmation primitives from the DLL and send them to the ARPM in a form of FAL internal primitives.

7 AP-Context state machine

7.1 Overview

The AP-Context State Machine (APCSM) manages the behavioral states, transitions and interactions of all the objects contained in an implementation of the Type 16 FAL. As shown in Figure 2, there are three states. Event notifications are delivered to the APCSM from the FAL Services Protocol Machine (FSPM) or the AR Protocol Machine (ARPM) as specified. These event notifications, identified in Figure 2, result in state transitions.

The APCSM is initiated in the Idle state.



Figure 2 – APCSM state diagram

7.2 States

7.2.1 Idle

7.2.1.1 Behavior

The FAL is not connected to the network. The only behavior associated with the Idle state is a determination of errors or other fault conditions that are specified as preventing the transition to the Running state. If there are no such preventative conditions, an automatic invocation of the Establish primitive is self generated by the APCSM.

7.2.1.2 Events

- **Establish** – Connects the FAL to the network. Upon successful completion, the APCSM transitions to the Running state.

7.2.2 Running

7.2.2.1 Behavior

The FAL is connected to the network. The Master type device is able to perform the behavior associated with communicating to the Slaves to which it is connected. Slave type devices are able to perform the behavior associated responding to the Master to which it is connected.

7.2.2.2 Events

- **Release** – Disconnects the FAL from the network. Upon successful completion, the APCSM transitions to the Idle state.

7.3 States, events and transitions

The combination of all states events and possible transitions are shown in Table 5.

Table 5 – APCSM state-event table

Current State	Event	Action	Next State
Idle	Establish	Initiate the methods specified for the Running state	Running
Idle	Release	Return an error	Running
Running	Establish	Return an error	Running
Running	Release	Initiate the methods specified for the Idle state	Idle

8 FAL service protocol machine (FSPM)

8.1 Overview

The FSPM provides the interface to the FAL user in the form of service handlers which convert service parameters into APDUs and process service requests from the FAL user or convert APDUs into service parameters and deliver service indications to the FAL user.

The FSPM operates in a single state with events defined by the receipt of service primitives.

8.2 MGT services

8.2.1 Get network status

Upon receipt of a Get Network Status service request from the FAL user, the FSPM prepares and delivers an appropriately encoded FSP-get network status primitive to the ARPM.

8.2.2 Get device status

Upon receipt of a Get Device Status service request from the FAL user, the FSPM prepares and delivers an appropriately encoded FSP-get device status primitive to the ARPM.

8.2.3 Network status change report

Upon receipt of an FSP-network status change indication from the ARPM, the FSPM prepares and delivers an appropriately encoded network status change indication to the FAL user.

8.2.4 Device status change report

Upon receipt of an FSP-device status change indication from the ARPM, the FSPM prepares and delivers an appropriately encoded device status change indication to the FAL user.

8.2.5 Set device status

Upon receipt of a Set Device Status service request from the FAL user, the FSPM prepares and delivers an appropriately encoded FSP-set device status primitive to the ARPM.

8.2.6 Enable RTC

Upon receipt of a Enable RTC service request from the FAL user, the FSPM prepares and delivers an appropriately encoded FSP-enable RTC primitive to the ARPM.

8.2.7 Notify RTC

Upon receipt of an FSP-notify RTC indication from the ARPM, the FSPM prepares and delivers an appropriately encoded notify RTC change indication to the FAL user.

8.2.8 Disable RTC

Upon receipt of a Disable RTC service request from the FAL user, the FSPM prepares and delivers an appropriately encoded FSP-disable RTC primitive to the ARPM.

8.2.9 Notify error

Upon receipt of an FSP-notify error indication from the ARPM, the FSPM prepares and delivers an appropriately encoded notify error indication to the FAL user.

8.3 IDN services

8.3.1 Read

Upon receipt of a Read service request from the FAL user, if the ARPM is not in the Running state, an error is returned to the FAL user; otherwise the FSPM prepares and delivers an appropriately encoded FSP-read primitive to the ARPM.

8.3.2 Write

Upon receipt of a Write service request from the FAL user, if the ARPM is not in the Running state, an error is returned to the FAL user; otherwise the FSPM prepares and delivers an appropriately encoded FSP-write primitive to the ARPM.

8.4 CYCIDN services

8.4.1 Read_cyclic

Upon receipt of a Read_cyclic service request from the FAL user, the FSPM prepares and delivers an appropriately encoded FSP-read cyclic primitive to the ARPM.

8.4.2 Write_cyclic

Upon receipt of a Write_cyclic service request from the FAL user, if the ARPM is not in the Running state, an error is returned to the FAL user, otherwise the FSPM prepares and delivers an appropriately encoded FSP-write cyclic primitive to the ARPM.

8.4.3 Notify_cyclic

Upon receipt of an FSP-notify cyclic indication from the ARPM, the FSPM prepares and delivers an appropriately encoded notify indication to the FAL user.

8.5 File transmission services

8.5.1 File_Download

Upon receipt of a File_Download service request from the FAL user if the ARPM is in the Running state, an error is returned to the FAL user; otherwise the FSPM prepares and delivers an appropriately encoded FSP-FD primitive to the ARPM.

8.5.2 File_Upload

Upon receipt of a File_Upload service request from the FAL user, if the ARPM is in the Running state, an error is returned to the FAL user; otherwise the FSPM prepares and delivers an appropriately encoded FSP-FU primitive to the ARPM.

9 Application relationship protocol machine (ARPM)

9.1 Overview

The ARPM manages the functions and behaviors of the ARs by

- a) receiving, decoding and processing service requests from the FSPM,
- b) preparing, encoding and delivering service requests to the DMPM,
- c) receiving, decoding and processing service indications from the DMPM,
- d) preparing, encoding and delivering service indications to the FSPM,
- e) monitoring critical functions of the ARs including timeout times and other fault conditions,
- f) delivering event notifications to the APCSM.

The behavior of the ARPM is managed by the APCSM.

There are two types of AR: Master and Slave.

9.2 Master ARPM

9.2.1 Overview

The Master ARPM manages the behavioral states, transitions and interactions of a Master AR. As shown in Figure 3, there are two states.

Sub-states are not represented as definitive states, but exist as abstractions used to identify a set of behaviors with a state.

The ARPM is initiated in the Idle state.

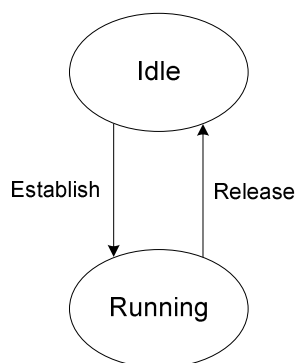


Figure 3 – ARPM master AR state diagram

9.2.2 State descriptions

9.2.2.1 Idle

9.2.2.1.1 Behavior

The FAL is not connected to the network. The only behavior associated with the Idle state is a determination of errors or other fault conditions that are specified as preventing the transition to the Running state.

9.2.2.1.2 Events

Establish – Connects the FAL to the network. Upon successful completion, the Master ARPM transitions to the Running state.

9.2.2.2 Running state

9.2.2.2.1 Behavior

In this state the FAL is connected to the network.

9.2.2.2.2 Events

Release – Disconnects the FAL from the network. Upon successful completion, the Master ARPM transitions to the Idle state.

9.2.3 States, events and transitions

The combination of all states events and possible transitions are shown in Table 6.

Table 6 – ARPM master state-event table

Current State	Event	Action	Next State
Idle	Establish	Initiate the methods specified for the Running state	Running
Idle	Release	Return an error	Running
Running	Establish	Return an error	Running
Running	Release	Initiate the methods specified for the Idle state	Idle

9.3 Slave ARPM

9.3.1 Overview

The Slave ARPM manages the behavioral states, transitions and interactions of a Slave AR. As shown in Figure 4, there are two states.

Sub-states are not represented as definitive states, but exist as abstractions used to identify a set of behaviors with a state.

The ARPM is initiated in the Idle state.

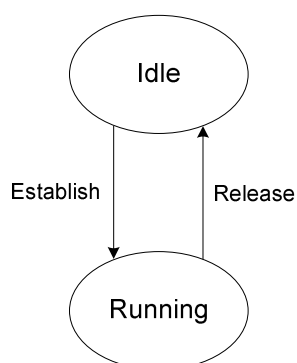


Figure 4 – ARPM slave AR state diagram

9.3.2 State descriptions

9.3.2.1 Idle

9.3.2.1.1 Behavior

The FAL is not connected to the network. The only behavior associated with the Idle state is a determination of errors or other fault conditions that are specified as preventing the transition to the Running state.

9.3.2.1.2 Events

Establish – Connects the FAL to the network. Upon successful completion, the Slave ARPM transitions to the Running state.

9.3.2.2 Running state

9.3.2.2.1 Behavior

In this state the FAL is connected to the network.

9.3.2.2.2 Events

Release – Disconnects the FAL from the network. Upon successful completion, the Slave ARPM transitions to the Idle state.

9.3.3 States, events and transitions

The combination of all states events and possible transitions are shown in Table 7.

Table 7 – ARPM slave state-event table

Current State	Event	Action	Next State
Idle	Establish	Initiate the methods specified for the Running state	Running
Idle	Release	Return an error	SAME
Running	Establish	Return an error	SAME
Running	Release	Initiate the methods specified for the Idle state	Idle

9.4 Primitives received from the FSPM

9.4.1 FSP-get network status

Upon receipt of an FSP-get network status service request from the FAL user, the ARPM prepares and delivers an appropriately encoded ARP-get network status primitive to the DMPM if the ARPM is in the running state. Otherwise, it discards the request. This primitive uses no attributes.

9.4.2 FSP-get device status

Upon receipt of an FSP-get device status service request from the FAL user, the ARPM prepares and delivers an appropriately encoded ARP-get device status primitive to the DMPM if the ARPM is in the running state. Otherwise, it discards the request. This primitive uses the network address of the selected device as attribute.

9.4.3 FSP-set device status

The FSP-set device status primitive is only valid for M type FAL users.

Upon receipt of an FSP-set device status from the FAL user, the APCSM prepares and delivers an appropriately encoded ARP-set device status primitive to the DMPM if the ARPM is in the running state. Otherwise, it discards the request. This primitive uses the network address of the selected device and the status to be set as attributes.

9.4.4 FSP-enable RTC

The FSP-enable RTC primitive is only valid for M type FAL users.

Upon receipt of an enable RTC service request from the FAL user, if the ARPM is in the Running state, an error is returned to the FAL user, otherwise the APCSM prepares and delivers an appropriately encoded ARP-enable RTC primitive to the DMPM. This primitive uses the network addresses of the selected devices as attributes.

9.4.5 FSP-disable RTC

The FSP-disable RTC primitive is only valid for M type FAL users.

Upon receipt of a disable RTC service request from the FAL user, if the ARPM is in the Idle state, an error is returned to the FAL user, otherwise the APCSM prepares and delivers an appropriately encoded ARP-disable RTC primitive to the DMPM. This primitive uses the network addresses of the selected devices as attributes.

9.4.6 FSP-read

The FSP-read primitive is only valid for M type FAL users.

Upon receipt of a read service request from the FAL user, the APCSM prepares and delivers an appropriately encoded ARP-read acyclic primitive to the DMPM if the ARPM is in the running state. Otherwise, it discards the request. This primitive uses the network addresses of the selected device and the IDN as attributes.

9.4.7 FSP-write

The FSP-write primitive is only valid for M type FAL users.

Upon receipt of a write service request from the FAL user, the APCSM prepares and delivers an appropriately encoded ARP-write acyclic primitive to the DMPM if the ARPM is in the running state. Otherwise, it discards the request. This primitive uses the network addresses of the selected device, the IDN and the value to be written as attributes.

9.4.8 FSP-read cyclic

Upon receipt of a read_cyclic service request from the FSPM, if the APCSM is not in the Running state, an error is returned; otherwise the ARPM prepares and delivers an appropriately encoded ARP-read cyclic primitive to the DMPM. This primitive uses the network addresses of the selected device and the IDN as attributes.

9.4.9 FSP-write cyclic

Upon receipt of a write_cyclic service request from the FSPM, if the APCSM is not in the Running state, an error is returned; otherwise the ARPM prepares and delivers an appropriately encoded ARP-write cyclic primitive to the DMPM. This primitive uses the network addresses of the selected device, the IDN and the value to be written as attributes.

9.4.10 FSP-FD

Upon receipt of a File_Download service request from the FSPM, if the APCSM is in the Running state, an error is returned; otherwise the ARPM prepares and delivers an appropriately encoded ARP-FD primitive to the DMPM. This primitive uses the network addresses of the selected device and the file to be downloaded as attributes.

9.4.11 FSP-FU

Upon receipt of a File_Upload service request from the FSPM, if the APCSM is in the Running state, an error is returned; otherwise the ARPM prepares and delivers an appropriately encoded ARP-FU primitive to the DMPM. This primitive uses the network addresses of the selected device and the file to be uploaded as attributes.

9.5 Indications received from the DMPM

9.5.1 ARP-network status change report

The ARP-network status change report indication is only valid for M type FAL users.

Upon receipt of an ARP-network status change report indication from the DMPM, the ARPM prepares and delivers an appropriately encoded FSP-network status change indication to the FSPM.

If the network status does not indicate an error, an Establish event notification is delivered to the APCSM.

If the network status does indicate an error, a Release event notification is delivered to the APCSM.

9.5.2 ARP-device status change report

Upon receipt of an ARP-device status change report indication from the DMPM, the ARPM prepares and delivers an appropriately encoded FSP-device status change indication to the FSPM.

9.5.3 ARP-notify RTC enabled

The ARP-notify RTC enabled indication is only valid for S type FAL users.

Upon receipt of an ARP-notify RTC enabled from the DMPM, the ARPM prepares and delivers an appropriately encoded FSP-notify RTC enabled to the FSPM.

An Establish event notification is delivered to the APCSM.

9.5.4 ARP-notify RTC disabled

The ARP-notify RTC disabled indication is only valid for S type FAL users.

Upon receipt of an ARP-notify RTC disabled from the DMPM, the ARPM prepares and delivers an appropriately encoded FSP-notify RTC disabled to the FSPM.

A Release event notification is delivered to the APCSM.

9.5.5 ARP-notify cyclic

Upon receipt of an ARP-notify cyclic from the DMPM, the ARPM prepares and delivers an appropriately encoded FSP-notify cyclic to the FSPM.

9.5.6 ARP-notify Error

Upon receipt of an ARP-notify Error from the DMPM, the ARPM prepares and delivers an appropriately encoded FSP-notify Error to the FSPM.

10 DLL mapping protocol machine (DMPM)

10.1 Overview

The DMPM maps the ARPM service requests to DL service requests (converting APDUs to DLSDUs) and DL service indications to ARPM service indications (converting DLSDUs to APDUs).

10.2 Primitives received from the ARPM

The mapping of ARPM primitives to DL service requests is specified in Table 8.

Table 8 – ARPM to DL mapping

ARPM primitive	DL service
ARP-read acyclic	Read (RD)
ARP-write acyclic	Write (WR)
ARP-enable RTC	Initiate_Cyclic_Communication (ICC)
ARP-read cyclic	Read_Cyclic (RDC)
ARP-write cyclic	Write_Cyclic (WRC)
ARP-disable RTC	Disable_Cyclic_Communication (DCC)
ARP-get network status	Get_Network_Status (GNS)
ARP-get device status	Get_Device_Status (GDS)
ARP-set device status	Set_Device_Status (SDS)
ARP-FD	File_Download (FD)
ARP-FU	File_Upload (FU)

10.3 Indications received from the DL

The mapping of DL service indications to ARPM indications is specified in Table 9.

Table 9 – DL to ARPM mapping

DL service indication	ARPM indication
Notify_Cyclic_Communication (NCC)	ARP-notify RTC enabled
Notify_Cyclic_Data (NCD)	ARP-notify cyclic
Notify_Error (NER)	ARP-notify Error
Notify_Cyclic_Communication_Disabled (NCCD)	ARP-notify RTC disabled
Notify_Network_Status_Change (NNSC)	ARP-network status change
Notify_Device_Status_Change (NDSC)	ARP-device status change

Bibliography

IEC 60559, *Binary floating-point arithmetic for microprocessor systems*

IEC/TR 61158-1 (Ed.2.0), *Industrial communication networks – Fieldbus specifications – Part 1: Overview and guidance for the IEC 61158 and IEC 61784 series*

IEC 61158-3-16, *Industrial communication networks – Fieldbus specifications – Part 3-16: Data-link layer service definition – Type 16 elements*

IEC 61784-1 (Ed.2.0), *Industrial communication networks – Profiles – Part 1: Fieldbus profiles*

ISO/IEC 646, *Information technology – ISO 7-bit coded character set for information interchange*

ISO/IEC 8825, *Information technology – ASN.1 encoding rules: Specification of Basic Encoding Rules (BER), Canonical Encoding Rules (CER) and Distinguished Encoding Rules (DER)*

ISO/IEC 8859-1, *Information technology – 8-bit single-octet coded graphic character sets – Part 1: Latin alphabet No. 1*

ISO/IEC 8877, *Information technology – Telecommunications and information exchange between systems – Interface connector and contact assignments for ISDN Basic Access Interface located at reference points S and T*

ISO 8601, *Data elements and interchange formats – Information interchange – Representation of dates and times*

ISO 15745-1:2003, *Industrial automation and systems integration – Open systems application integration framework – Part 4: Reference description for Ethernet-based control systems, Amendment 1 (2007): PROFINET profiles*

INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

3, rue de Varembé
P.O. Box 131
CH-1211 Geneva 20
Switzerland

Tel: + 41 22 919 02 11
Fax: + 41 22 919 03 00
info@iec.ch
www.iec.ch